Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 3

Title: Automating Retail Performance Reporting: A Power BI Dashboard for Sales, Revenue, and Profit Analysis

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Automating Retail Performance Reporting: A Power BI Dashboard for Sales, Revenue, and Profit Analysis

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Problem Statement

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A retail company currently relies on manually prepared reports to monitor its business performance, a process that is time-consuming, error-prone, and unable to provide real-time visibility into sales, revenue, profit, and product performance; as transactional data grows across products, stores, and customers, this manual, spreadsheet-based approach becomes inefficient for identifying trends and making timely decisions, creating a need for an interactive Power BI dashboard that consolidates the data, calculates key performance metrics, and visually presents sales, revenue, profit, and product-wise insights to enable faster, data-driven decision-making.

Methodology / Steps Followed

Step 1: Data Collection

- Retail transactional data sourced from the Kaggle "Retail Sales Dataset," organized into separate tables: Sales, Customer, Store, and Product
- Product master data (Category, ProductID, ProductName, SubCategory) stored separately in a MySQL database (retail_db)
- Customer data collected with fields such as Customer ID, Customer Name, and Segment
- Store data collected with Region, State, and StoreID to support geographic analysis
- Sales data includes Order Date, Ship Date, Discount, Profit, Quantity, Sales, and Ship Mode at the transaction level

Step 2: Data Import

- All flat files (Customer, Sales, Store, Product, Time) imported into Power BI using the "Get Data" → Excel/CSV workbook connector
- All tables loaded into Power BI Desktop's data model through Power Query

- Verified successful load of each table by checking row counts and column headers post-import
- Reviewed each table's preview in Power Query Editor before loading to catch any formatting issues early

Step 3: Data Cleaning and Transformation

- Checked and corrected data types for each column (dates as Date, Sales/Profit/Discount as Decimal Number, IDs as Text/Whole Number)
- Removed duplicate rows and blank/null entries from the Sales dataset using Power Query
- Standardized column names across tables for consistency (e.g., Customer ID, Product ID, Store ID)
- Verified referential integrity between keys (Customer ID, Product ID, StoreID) across tables before modeling relationships

Step 4: Data Modeling

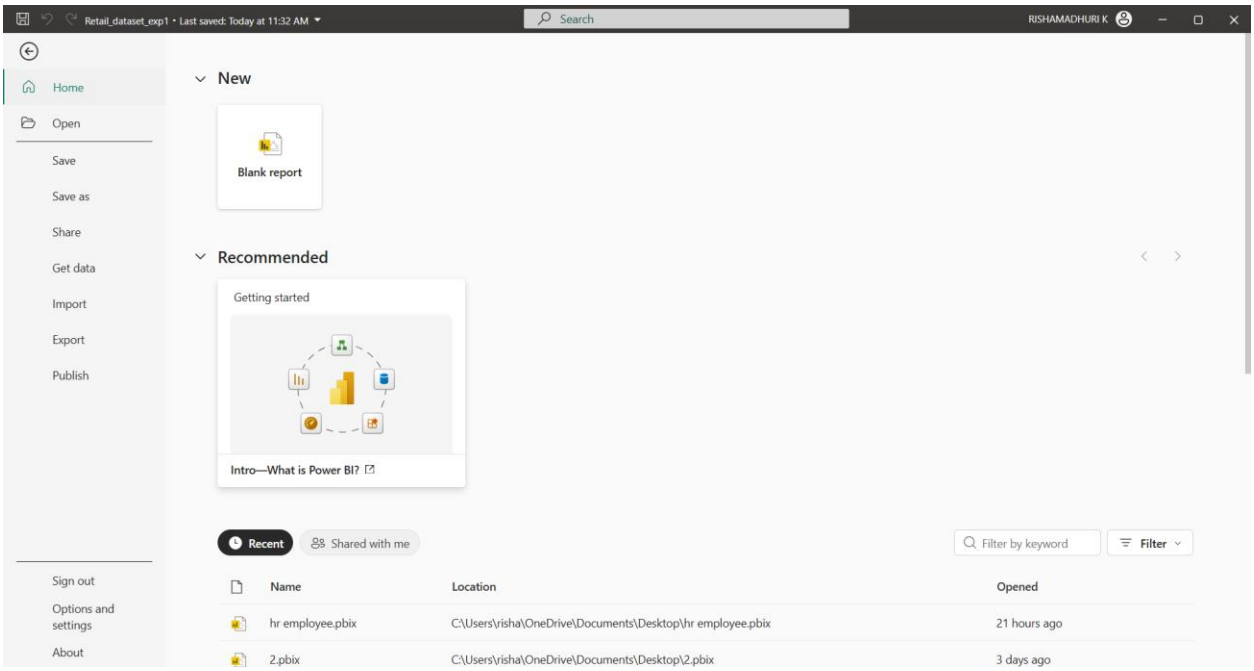
- Built a star-schema-style data model connecting Sales_dataset (fact table) to Customer_Data, Product, Store_Data, and Time_Data (dimension tables) via one-to-many relationships
- Created relationships on Customer ID, Product ID, StoreID, and Date fields to enable cross-filtering
- Wrote DAX measures: Total Sales, Total Profit, Total Orders, Total Quantity, Total Customers, Profit Margin %, Revenue per Order, and Units per Order
- Validated relationships by checking that filters applied on one table correctly cross-filtered related tables

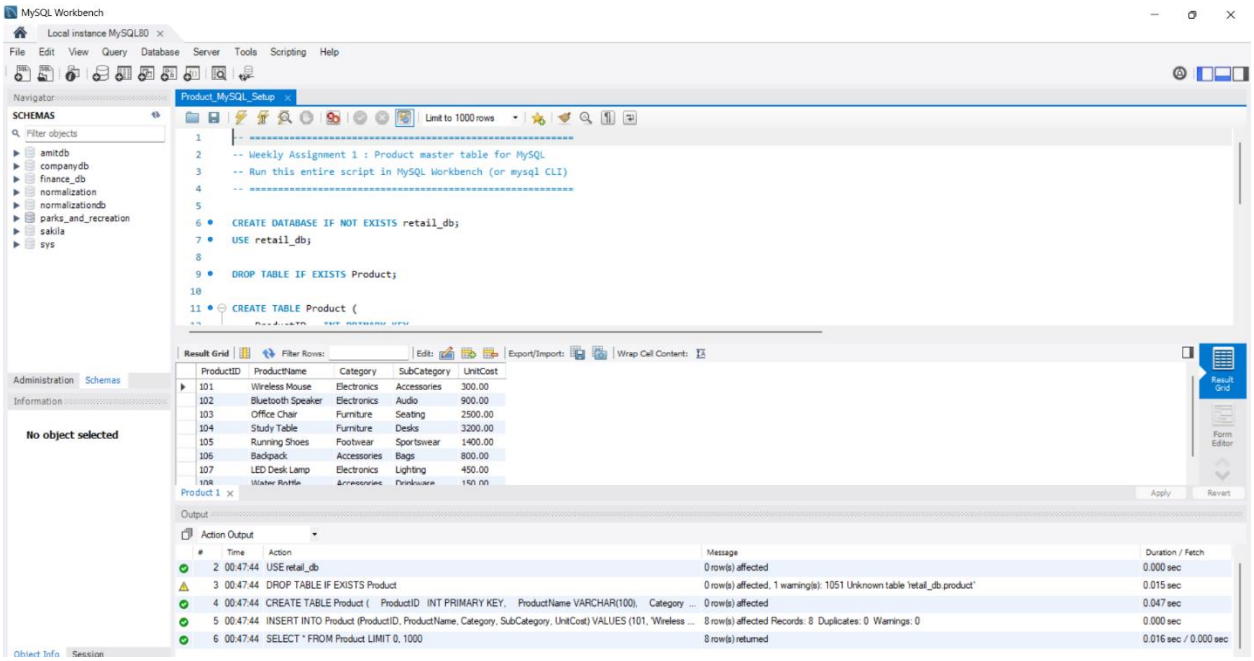
Step 5: Dashboard Development

- Designed KPI cards for Total Sales, Total Profit, Profit Margin %, Total Orders, and Revenue per Order
- Built visuals: Sales & Profit by State (map), Discount Impact on Profitability (scatter), Total Sales by MonthName (line/area)
- Added Regional Profitability by Category (matrix) and Segment Sales & Profitability (combo chart) for deeper comparisons

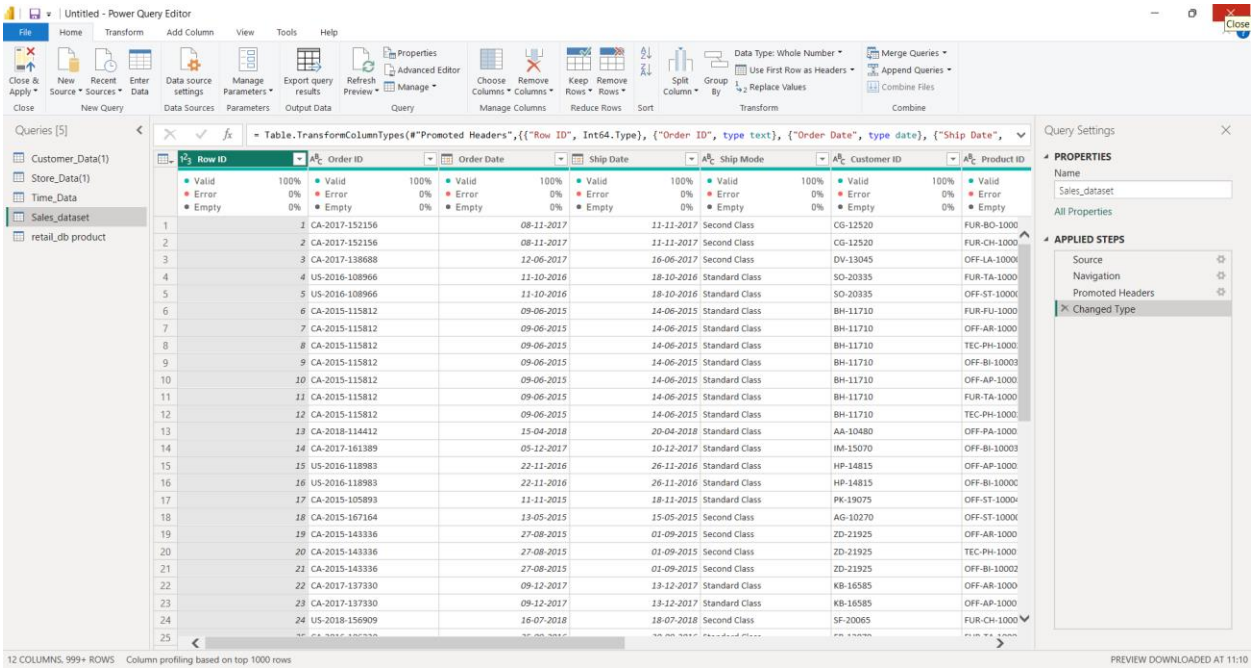
- Arranged all visuals on a single dashboard page for a consolidated, at-a-glance view of business performance

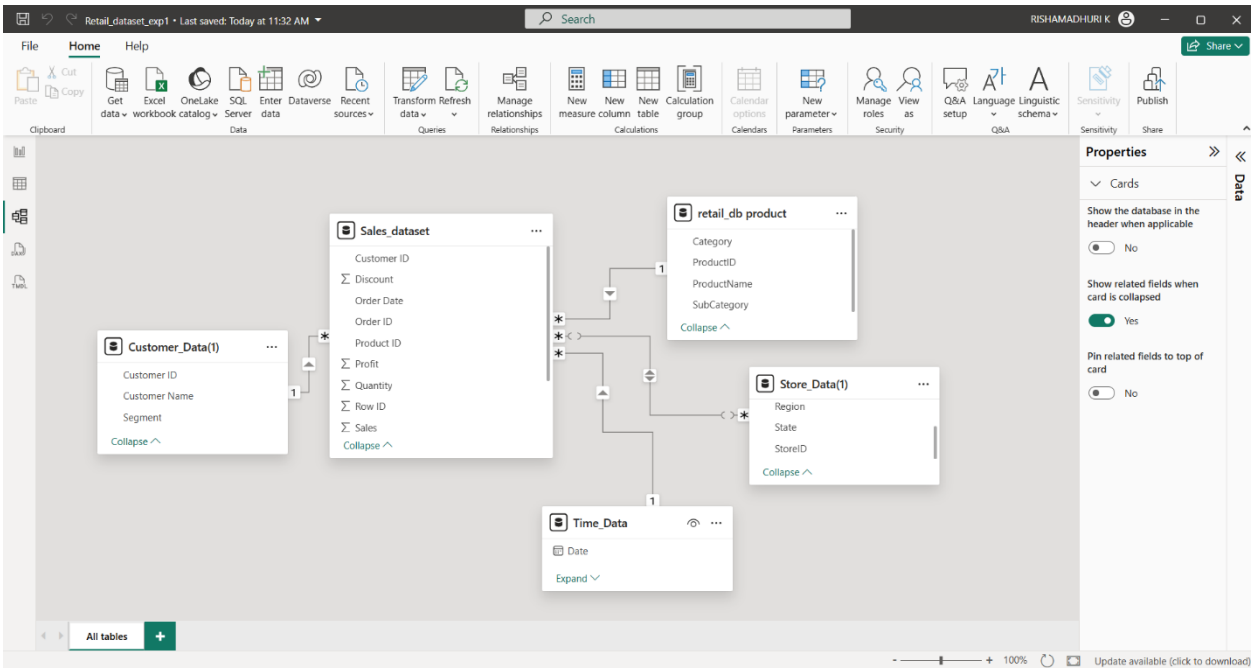
Output

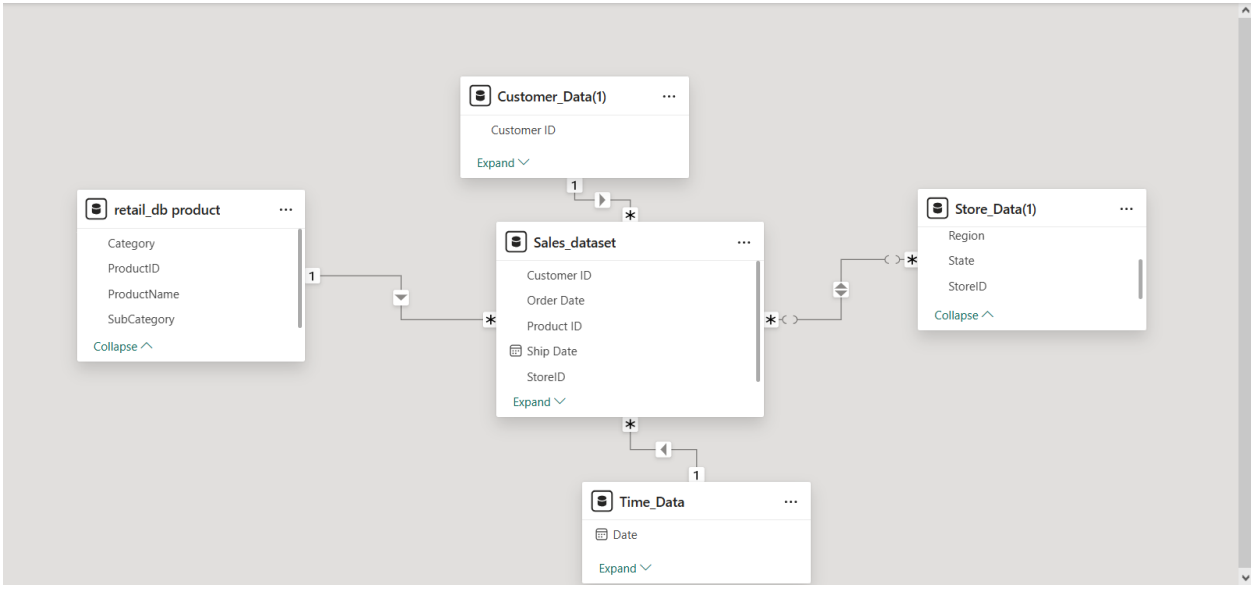


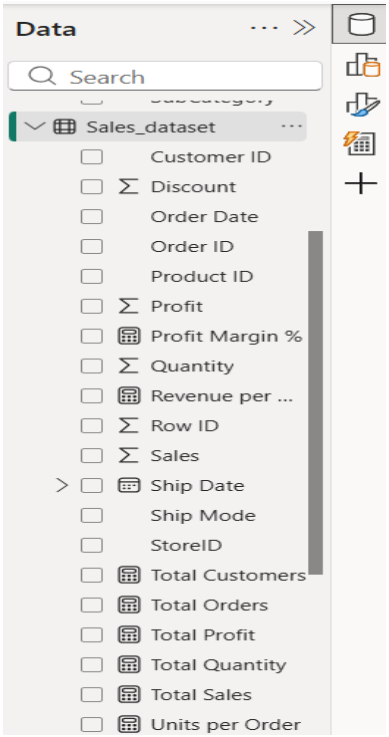


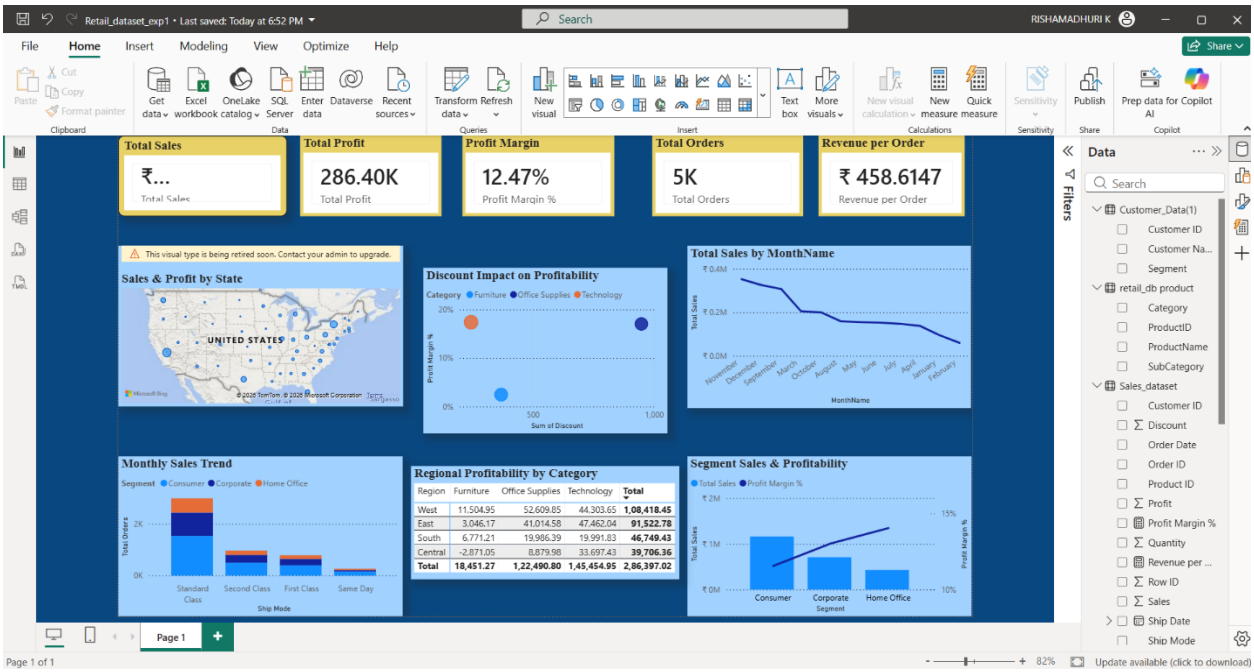

MySQL database
Server
127.0.0.1:3306
Database
retail_db
Advanced options
OK
Cancel


Untitled - Power Query Editor
Queries [5]
Customer_Data(1)
Store_Data(1)
Time_Data
Sales_dataset
retail_db product
Query Settings
PROPERTIES
Name
Sales_dataset
All Properties
APPLIED STEPS
Source
Navigation
Promoted Headers
Changed Type


Retail_dataset_exp1 • Last saved: Today at 11:32 AM
Search
RISHAMADHURI K
File
Home
Help
Share
Properties
Cards
Data
Sales_dataset
Customer ID
Discount
Order Date
Order ID
Product ID
Profit
Quantity
Row ID
Sales
Collapse
retail_db product
Category
ProductID
ProductName
SubCategory
Collapse
Customer_Data(1)
Customer ID
Customer Name
Segment
Collapse
Store_Data(1)
Region
State
StoreID
Collapse
Time_Data
Date
Expand
All tables
Show the database in the header when applicable
No
Show related fields when card is collapsed
Yes
Pin related fields to top of card
No
100%
Update available (click to download)


Customer_Data(1)
Customer ID
Expand
retail_db product
Category
ProductID
ProductName
SubCategory
Collapse
Sales_dataset
Customer ID
Order Date
Product ID
Ship Date
StoreID
Expand
Store_Data(1)
Region
State
StoreID
Collapse
Time_Data
Date
Expand


Data
Search
Sales_dataset
Customer ID
Discount
Order Date
Order ID
Product ID
Profit
Profit Margin %
Quantity
Revenue per ...
Row ID
Sales
Ship Date
Ship Mode
StoreID
Total Customers
Total Orders
Total Profit
Total Quantity
Total Sales
Units per Order


Retail_dataset_exp1 • Last saved: Today at 6:52 PM
Search
RISHAMADHURI K
File
Home
Insert
Modeling
View
Optimize
Help
Share
Total Sales
₹...
Total Profit
286.40K
Total Profit
Profit Margin
12.47%
Profit Margin %
Total Orders
5K
Total Orders
Revenue per Order
₹ 458.6147
Revenue per Order
This visual type is being retired soon. Contact your admin to upgrade.
Sales & Profit by State
UNITED STATES
Discount Impact on Profitability
Total Sales by MonthName
Monthly Sales Trend
Regional Profitability by Category
Region Furniture Office Supplies Technology Total
West 11,504.95 52,609.85 44,303.65 1,08,418.45
East 3,046.17 41,014.58 47,462.04 91,522.78
South 6,771.21 19,986.39 19,991.83 46,749.43
Central -2,871.05 8,879.98 33,697.43 39,706.36
Total 18,451.27 1,22,490.80 1,45,454.95 2,86,397.02
Segment Sales & Profitability
Filters
Data
Search
Customer_Data(1)
retail_db product
Sales_dataset
Page 1
Page 1 of 1
Update available (click to download)

- A single-page interactive Power BI dashboard displaying Total Sales, Total Profit (₹286.40K), Profit Margin (12.47%), Total Orders (5K), and Revenue per Order (₹458.61)
- Geographic visualization showing sales and profit distribution across U.S. states
- Category-wise profitability breakdown showing how discount levels impact profit margin across Furniture, Office Supplies, and Technology
- Monthly sales trend line highlighting seasonal patterns in overall sales
- Regional performance matrix comparing Central, East, South, and West regions across product categories
- Segment-wise comparison (Consumer, Corporate, Home Office) showing both sales volume and profitability side by side

Conclusion

This project successfully replaced manual, spreadsheet-based reporting with an automated, interactive Power BI dashboard by consolidating multiple related datasets—Sales, Customer, Product, Store, and Time—into a single, unified data model. The resulting single-page dashboard presents sales, revenue, profit, and product performance in one consolidated view, giving management a real-time tool for tracking key performance indicators and significantly reducing the time and effort required compared to manual methods, ultimately meeting the objective of transforming static reports into a dynamic decision-support system.

Learning Outcomes

- Gained hands-on experience importing and integrating multiple related flat-file tables into a single Power BI data model
- Learned to design relational data models using fact and dimension tables with proper one-to-many relationships
- Developed practical skills in writing DAX measures for business KPIs such as Profit Margin %, Revenue per Order, and Total Orders
- Learned to select and design appropriate visuals (maps, scatter plots, matrices, combo charts) for different types of business questions

Business Insights

- Certain regions (e.g., West and East) contribute a significantly higher share of total sales and profit compared to others, indicating where sales efforts are strongest
- Higher discount levels are associated with lower profit margins in categories like Furniture, suggesting discounting strategy needs review for that category
- Monthly sales trends reveal seasonal peaks and dips, which can help with inventory

planning and targeted marketing timing

- The Consumer segment generates the highest sales volume, but Corporate and Home Office segments show notable profitability, indicating a case for segment-specific pricing or service strategies